Machine Learning and Pattern Recognition

Introduction

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February 14, 2025

Introduction

- Introduction
- An Example of Supervised Learning
- Introduction to Probability Theory
- Introduction to Decision Theory
- Introduction to Information Theory

Outline

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Introduction

Benefits of using GitHub

- Version Control: Tracking changes in the code over time.
- Code Sharing: Sharing projects with others.
- Portfolio Building: Build a portfolio of your work (internship application)
- Learning and Growth: GitHub is widely used in the industry, so getting familiar with it early on can give you a head start in your career

If you are not sure what to share, that brings us to Lecture 2.

Introduction

In the first class, we have studied,

- What is Machine Learning?
- Deep Learning
- Artificial Intelligence (AI)
- Neural Networks
- Machine Learning Tasks
- Training, Test, and Validation

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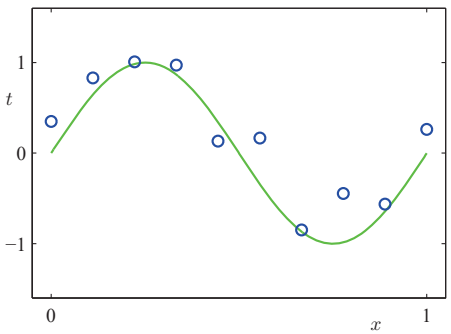
An Example of Supervised Learning



Training data of t_n and x_n

- We will start with a supervised learning problem
- Suppose we have N observations of x (time):

$$\mathbf{x} = (x_1, x_2, \ldots, x_N)^{\mathrm{T}}$$

- and N observations of t (temperature):

$$\mathbf{t} = (t_1, t_2, \ldots, t_N)^{\mathrm{T}}$$

- Task: find a ML model to compute $\widehat{t}$ for new values of $\widehat{x}$ by using the training data (blue circles)

An Example of Supervised Learning

- This is a typical supervised learning problem
- We are given pairs of input (x) and output (t) data for training
- We search for a machine learning model to characterize the relationship between input and output.

- Once we found the ML model, we can predict the output for new input data

An Example of Supervised Learning

- Consider a polynomial function

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \cdots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

- The coefficient vector: $\mathbf{w} = [w_0, w_1, \ldots, w_M]$
- To find coefficients $\mathbf{w}$, we fit the polynomial function to the training data. So we minimize the error function

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \left(y(x_n, \mathbf{w}) - t_n \right)^2$$

- This becomes a curve fitting problem. We denote the solution by $\mathbf{w}^\star$
- Still need to choose N in $y(x, \mathbf{w})$

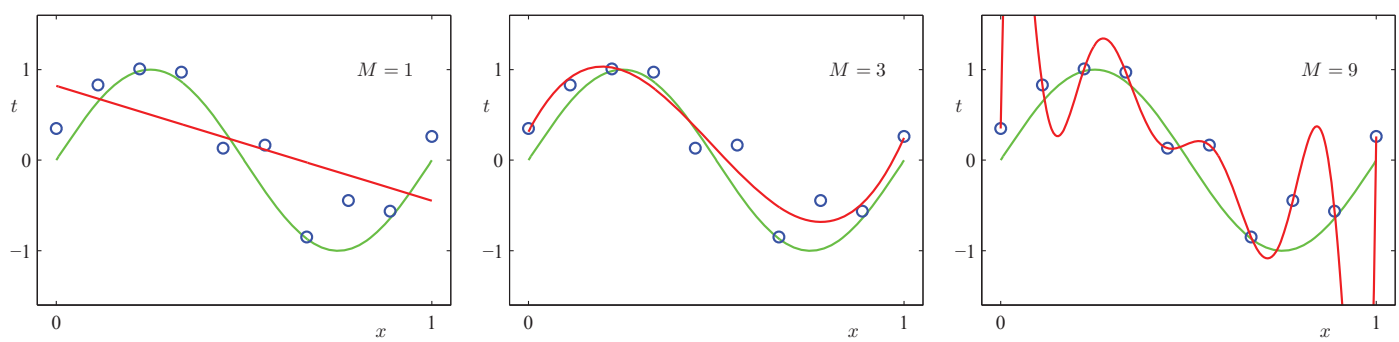
An Example of Supervised Learning

```python
# Python Program for Polynomial Fitting
# Load required lib
import numpy as np
import matplotlib.pyplot as plt
# Generate synthetic data by using the sine function with noise
def generate_noisy_sine_data(num_points=10, noise_level=0.2):
    x = np.linspace(0, 2 * np.pi, num_points)
    y = np.sin(x) + noise_level * np.random.randn(num_points)
    return x, y
# define a function to fit the data to the model (aka training)
def fit_polynomial(x, y, n):
    coeffs = np.polyfit(x, y, n)
    poly = np.poly1d(coeffs)
    print("Coefficients (descending order of powers):", coeffs)
    return poly
```

An Example of Supervised Learning

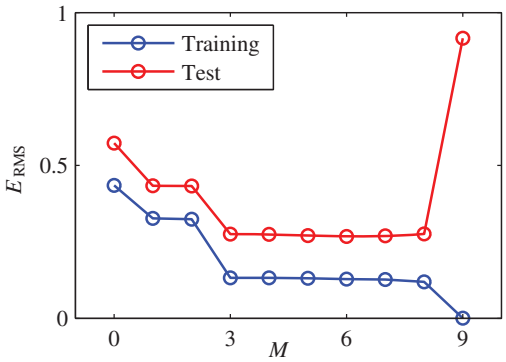
```python
# ... Continue from previous page ...
# define a function to plot the result
def plot_fitting(x, y, poly):
    ...

if __name__ == "__main__":
    np.random.seed(42)
    x, y = generate_noisy_sine_data()
    n = 3 # Change this value to adjust the polynomial order
    poly = fit_polynomial(x, y, n)
    plot_fitting(x, y, poly)
```


An Example of Supervised Learning



- Red curve: polynomial function with different models (order)
- $M = 1$: under fitting
- $M = 3$: best fit
- Higher-order ($M = 9$): over fitting

An Example of Supervised Learning



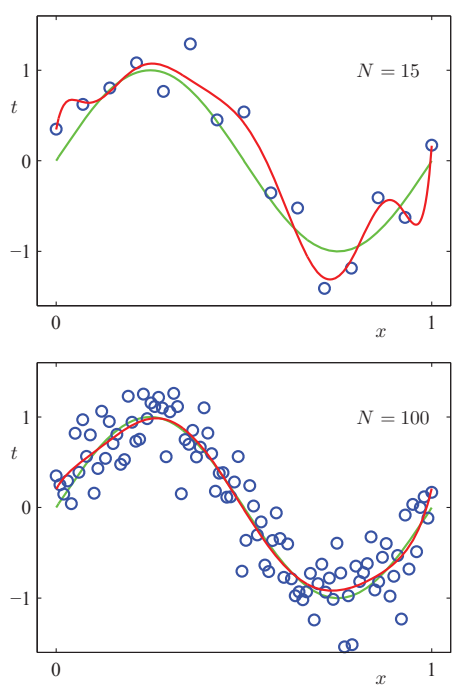
	$M=0$	$M=1$	$M=3$	$M=9$
$w_0^\star$	0.19	0.82	0.31	0.35
$w_1^\star$		-1.27	7.99	232.37
$w_2^\star$			-25.43	-5321.83
$w_3^\star$			17.37	48568.31
$w_4^\star$				-231639.30
$w_5^\star$				640042.26
$w_6^\star$				-1061800.52
$w_7^\star$				1042400.18
$w_8^\star$				-557682.99
$w_9^\star$				125201.43

- Considering a test set of $N = 100$ new data points generated the same way
- Using root-mean-square error function

$$E_{\mathrm{RMS}} = \sqrt{2E(\mathbf{w}^\star)/N}$$

- As shown, smaller M yields large error due to inflexible
- $3 \leq M \leq 8$: reasonable fit, small error
- $M = 9$: training set error is zero (?), test set error is large
- Coefficients gets larger as M increases

An Example of Supervised Learning



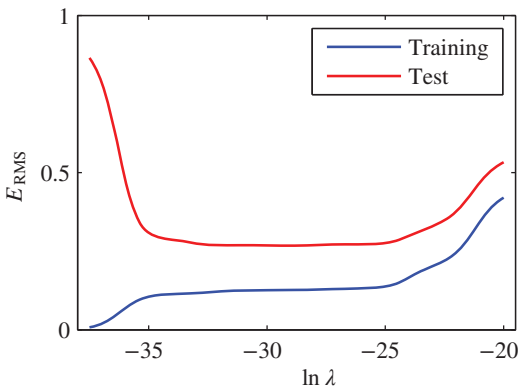
- For $M = 9$, error is reduced when N becomes larger
- For a good fit, the size of data set should be multiple of the number of parameters
- To control over-fitting, we can add a penalty term (regularization) and then minimize

$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(y(x_n, \mathbf{w}) - t_n\right)^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

while $\|\mathbf{w}\|^2 = \mathbf{w}^{\mathrm{T}}\mathbf{w}$. Often, w_0 is omitted

An Example of Supervised Learning

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
$w_0^\star$	0.35	0.35	0.13
$w_1^\star$	232.37	4.74	-0.05
$w_2^\star$	-5321.83	-0.77	-0.06
$w_3^\star$	48568.31	-31.97	-0.05
$w_4^\star$	-231639.30	-3.89	-0.03
$w_5^\star$	640042.26	55.28	-0.02
$w_6^\star$	-1061800.52	41.32	-0.01
$w_7^\star$	1042400.18	-45.95	-0.00
$w_8^\star$	-557682.99	-91.53	0.00
$w_9^\star$	125201.43	72.68	0.01



- With regularization, coefficients $\mathbf{w}$ are reduced
- As shown in the figure, λ controls the degree of over-fitting
- This curve fitting example seems to be intuition
- Next, we seek a more principled approach with the help of probability theory

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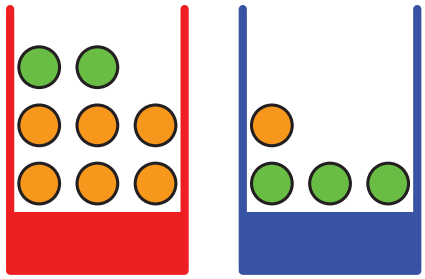
Introduction to Probability Theory

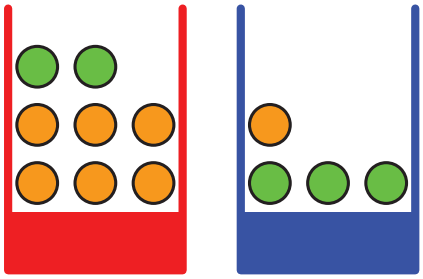
- Uncertainty arises both through noise on measurements and through the finite size of data sets
- Probability theory provides a consistent framework for the quantification and manipulation of uncertainty
- When combined with decision theory, it allows us to make optimal predictions given all the available information

Introduction to Probability Theory



- Imagine we have two boxes: red and blue
- We have 2 apples and 6 oranges in the red box, 3 apples and 1 orange in the blue one
- Now we randomly pick a box and then pick a fruit from that box.
- We observed the item and the color of the box, then we return it to the original box
- We repeat this process many times

Introduction to Probability Theory



- In this example, the identity of the box is a random variable (B), it can be: r or b

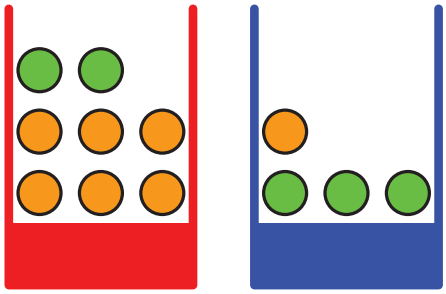
$$B = r, \quad B = b$$

- The identity of the fruit is also a random variable (F), it can be: a or o

$$F = a, \quad F = o$$

- Probability of an event: the fraction of the times that event occurs out of the total number of trials

Introduction to Probability Theory



- After many trials, the red box was picked 40% of the time, the blue box 60%, then,

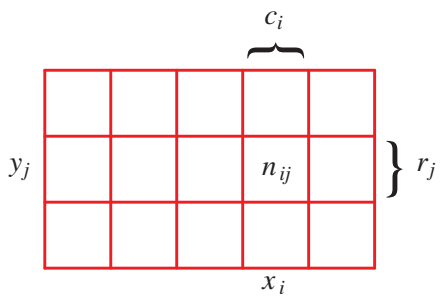
$$p(B = r) = \frac{4}{10}, \quad p(B = b) = \frac{6}{10}$$

- By definition, $p \in [0, 1]$ and if events are mutually exclusive and all possibilities are included,

$$\sum p_i = 1$$

- If we have chosen a apple, what is the probability that it came from the blue box?

Introduction to Probability Theory



$M = 5, L = 3$

- Consider a general example involving two random variables X and Y, and

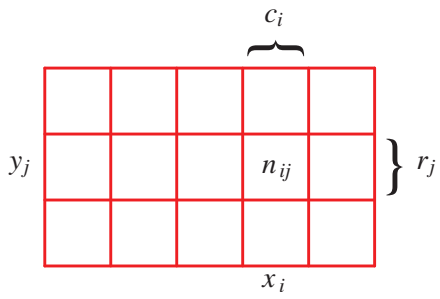
$$X \in \{x_i | i = 1, 2, \ldots, M\},$$
$$Y \in \{y_j | j = 1, 2, \ldots, L\}$$

- Suppose we performed N trials and the number of trials that $X = x_i$ and $Y = y_j$ is n_{ij}, then, joint probability of $X = x_i$ and $Y = y_j$ is

$$p(X = x_i, Y = y_j) = n_{ij}/N$$

- Let the number of trials that $X = x_i$ (irrespective of the value of Y) be c_i, the probability that $X = x_i$ is $p(X = x_i) = c_i/N$

Introduction to Probability Theory



$M = 5, L = 3$

- For trials that $X = x_i$, Y can be any value. So,

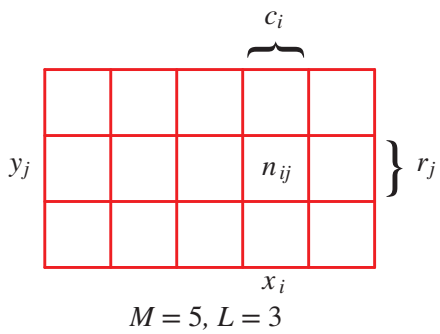
$$c_i = \sum_{j=1}^{L} n_{ij}$$

- Thus, we obtain the sum rule of probability as

$$p(X = x_i) = \sum_{j=1}^{L} p(X = x_i, Y = y_j)$$

- Note that $p(X = x_i)$ is called the marginal probability , because is is obtained by marginalizing.

Introduction to Probability Theory



- If we consider only the c_i trials that $X = x_i$, the fraction of trials that $Y = y_i$ is called conditional probability

$$p(Y = y_i | X = x_i) = n_{ij}/c_i$$

- With that, we obtain the product rule of probability as

$$p(X = x_i, Y = y_i) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i}\frac{c_i}{N}$$
$$= p(Y = y_i | X = x_i)p(X = x_i)$$

Introduction to Probability Theory

- With a more compact notation, we write

$$\text{sum rule: } p(X) = \sum_Y p(X, Y)$$
$$\text{product rule: } p(X, Y) = p(Y|X)p(X)$$

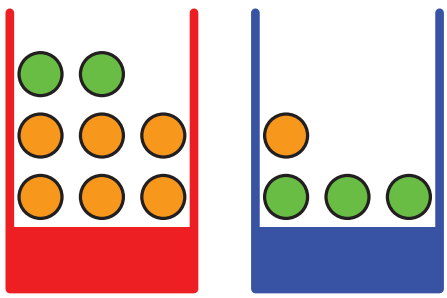
- From the product rule and the symmetry property $p(X, Y) = p(Y, X)$, we obtain Bayes' theorem as

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)} \quad \text{or} \quad p(Y|X)p(X) = p(X|Y)p(Y)$$

- Using the sum rule, we obtain

$$p(X) = \sum_Y p(X|Y)p(Y)$$

Introduction to Probability Theory



- Now if we pick a box randomly, and it turns out to be the blue box. Then, the probability of selecting an apple is

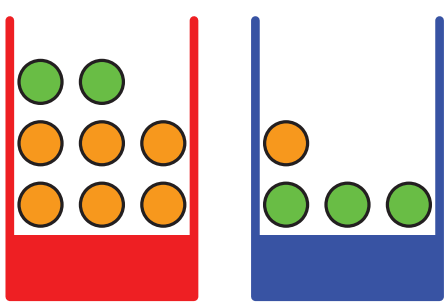
$$p(F = a|B = b) = 3/4$$

- Similarly, we obtain

$$p(F = o|B = b) = 1/4$$
$$p(F = a|B = r) = 1/4$$
$$p(F = o|B = r) = 3/4$$

- If we have chosen a orange, what is the probability that it came from the red box?

Introduction to Probability Theory



- The overall probability of picking an apple (sum and product rule)

$$\begin{aligned} p(F = a) &= p(F = a|B = r)p(B = r) \\ &+ p(F = a|B = b)p(B = b) \\ &= 11/20 \end{aligned}$$

- From which, we can find $p(F = o) = 9/20$
- Now if we have chosen a orange, what is the probability that it came from the red box?

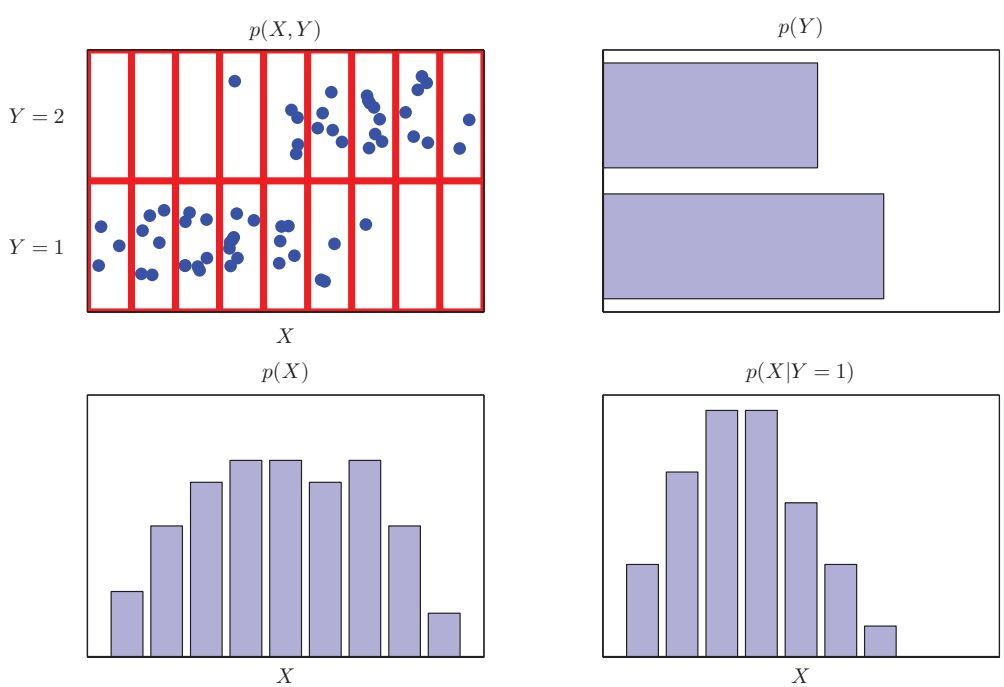
$$\begin{aligned} p(B = r|F = o) &= \frac{p(F = o|B = r)p(B = r)}{p(F = o)} \\ &= 2/3 \end{aligned}$$

Introduction to Probability Theory

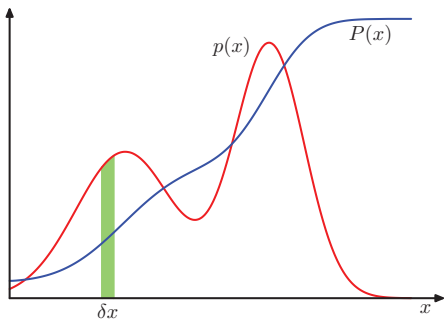
- If we had been asked the identity of box without knowing the identity of the selected fruit, then the most complete information we have is $p(B)$. We call this prior probability
- Once we know the identity of the fruit (orange), we can then use Bayes' theorem to compute the probability $p(B|F)$, which we call the posterior probability
- If the joint distribution of two variables factorizes into the product of the marginals, $p(X, Y) = p(X)p(Y)$, then X and Y are said to be independent

Introduction to Probability Theory

Now, consider a joint distribution of two variables: X and Y. We have $N = 60$ data points



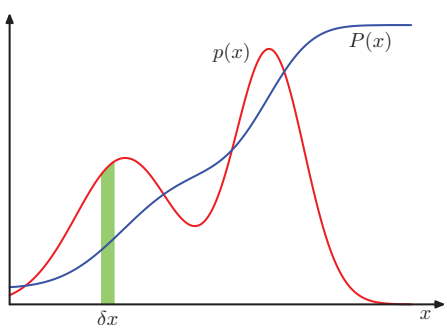
Introduction to Probability Theory



- Consider probabilities of real-valued continuous variables (x)
- If the probability of x falling in the interval $(x, x + \delta x)$ is given by $p(x)\delta x$, then $p(x)$ is called the probability density over x when $\delta x \to 0$
- The probability that x will lie in an interval (a, b) is

$$p(x \in (a, b)) = \int_a^b p(x)\,\mathrm{d}x$$

Introduction to Probability Theory



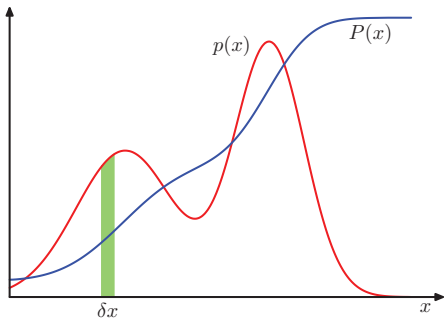
- The probability density $p(x)$ must satisfy

$$p(x) \geq 0$$
$$\int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1$$

- The probability that x lies in the interval $(-\infty, z)$ is called the cumulative distribution function

$$P(z) = \int_{-\infty}^{z} p(x)\,\mathrm{d}x$$

Introduction to Probability Theory



- The sum and product rules of probability, as well as Bayes' theorem, apply equally to the case of probability densities
- If x and y are two real variables, then

$$p(x) = \int p(x, y)\,\mathrm{d}y$$
$$p(x, y) = p(y|x)p(x)$$

Introduction to Probability Theory

- One of the most important tasks in probability theory is calculating the weighted averages of functions
- A random variable is a function mapping that assigns a real number $X(\omega)$ to each possible outcome (ω) of a random experiment
- For example, let's flip a coin 10 times (a random experiment)
- $X(\omega)$: the number of Heads we observe. If

$$\omega = HHTHTHHTHT \implies X(\omega) = 6$$

- We often shorten this and write the random variable as X

Introduction to Probability Theory

- From the coin example, we see that X does not have a fixed value, but rather a distribution of values it can take.
- We may ask what is the average number of heads in 10 experiments
- This average value is called the expectation, denoted as $\mathbb{E}[X]$, it is defined as

$$\mathbb{E}[X] = \sum_{a \in A} P(X = a)a$$

- "A" represents the set of all possible values X can take

Introduction to Probability Theory

- Suppose the random variable X represents the outcome of rolling a fair 6-sided die
- The possible values of X are: $X \in [1, 2, 3, 4, 5, 6]$
- Since the die is fair, each outcome has an equal probability: $P(X = x) = 1/6$
- The expectation of X is:

$$\begin{aligned}\mathbb{E}[X] &= \sum_{n=1}^{N} P(X = x_n)x_n \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}\end{aligned}$$

Introduction to Probability Theory

There are two important properties regarding expectations:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

This holds even if the random variables are dependent

$$\mathbb{E}[cX] = c\mathbb{E}[X],$$

where c is a constant

Note we do not have similar property for $\mathbb{E}[XY]$

Introduction to Probability Theory

- What is the expected value of the sum of two dice?
- X = value of roll 1
- Y = value of roll 2

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] = \frac{7}{2} + \frac{7}{2} = 7$$

- Comparing this to computing $2 \times \frac{1}{36} + 3 \times \frac{2}{36} + \cdots + 12 \times \frac{1}{36} = 7$

Introduction to Probability Theory

- Suppose a bakery sells two types of pastries: Croissants and Muffins
- The number of croissants sold per day is a random variable X, with an expected value $\mathbb{E}[X] = 30$
- The number of muffins sold per day is a random variable Y, with an expected value $\mathbb{E}[Y] = 20$
- We define a new random variable $Z = X + Y$, representing the total number of pastries sold per day.
- By the linearity of expectation:

$$\begin{aligned}\mathbb{E}[Z] &= \mathbb{E}[X + Y] \\ &= \mathbb{E}[X] + \mathbb{E}[Y] = 30 + 20 = 50\end{aligned}$$

Introduction to Probability Theory

- The average value of a function $f(x)$ under a probability distribution $p(x)$ is called the expectation of $f(x)$, denoted by $\mathbb{E}[f]$
- For a discrete distribution, it is given by

$$\mathbb{E}[f] = \sum_{n=1}^{N} p(x_n) f(x_n)$$

- In the case of continuous variables x

$$\mathbb{E}[f] = \int p(x) f(x) \, \mathrm{d}x$$

Introduction to Probability Theory

- The expectation only tells us the mean. Sometimes we also need to know how far away the data is distributed from the mean
- The variance for a random variable X with $\mathbb{E}[X] = \mu$ is defined as:

$$\mathrm{var}[x] = \mathbb{E}\left[(X-\mu)^2\right]$$

- The variance can be simplified as:

$$\begin{aligned}\mathbb{E}\left[(X-\mu)^2\right] &= \mathbb{E}\left(X^2 - 2\mu X + \mu^2\right)\\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2]\\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2\\ &= \mathbb{E}[X^2] - \mu^2\end{aligned}$$

Introduction to Probability Theory

- The variance of a function $f(x)$ is defined by

$$\mathrm{var}[f] = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]$$

  which provides a measure of how much variability there is in $f(x)$ around its mean value $\mathbb{E}[f(x)]$
- Expanding out the square, we obtain

$$\begin{aligned}\mathrm{var}[f] &= \mathbb{E}\left[f(x)^2 - 2f(x)\mathbb{E}[f(x)] + \mathbb{E}[f(x)]^2\right]\\ &= \mathbb{E}[(f(x)^2] - \mathbb{E}[f(x)]^2\end{aligned}$$

- In particular, if $f(x) = x$, then

$$\mathrm{var}[x] = \mathbb{E}\left[(x - \mathbb{E}[x])^2\right] = \mathbb{E}[x^2] - \mathbb{E}[x]^2$$

Introduction to Probability Theory

- Constants are squared when moving out:

$$\mathrm{var}[cX] = c^2 \mathrm{var}\,[X]$$

- For independent random variables X and Y , we have

$$\mathbb{E}\,[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

- and

$$\mathrm{var}[X + Y] = \mathrm{var}[X] + \mathrm{var}[Y]$$

Introduction to Probability Theory

- Consider expectation of functions of several variables, we use a subscript to indicate which variable is being averaged over

$$\mathbb{E}_x[f(x, y)]$$

  which becomes a function of y

- We can also consider a conditional expectation with respect to a conditional distribution

$$\mathbb{E}_x[f|y] = \sum_x p(x|y) f(x)$$

Introduction to Probability Theory

- For two random variables x and y, the covariance is defined by

$$\begin{aligned}
\operatorname{cov}[x, y] &= \mathbb{E}_{x,y}\left[(x - \mathbb{E}[x])\,(y - \mathbb{E}[y])\right] \\
&= \mathbb{E}_{x,y}\left[xy - x\mathbb{E}[y] - y\mathbb{E}[x] + \mathbb{E}[x]\mathbb{E}[y]\right] \\
&= \mathbb{E}_{x,y}[xy] - \mathbb{E}_{x,y}\left[x\mathbb{E}[y]\right] - \mathbb{E}_{x,y}\left[y\mathbb{E}[x]\right] + \mathbb{E}[x]\mathbb{E}[y] \\
&= \mathbb{E}_{x,y}[xy] - \mathbb{E}_{x,y}[x]\,\mathbb{E}[y] - \mathbb{E}_{x,y}[y]\,\mathbb{E}[x] + \mathbb{E}[x]\mathbb{E}[y] \\
&= \mathbb{E}_{x,y}[xy] - \mathbb{E}[x]\mathbb{E}[y]
\end{aligned}$$

  which expresses the extent to which x and y vary together.
- If x and y are independent, then their covariance vanishes

Introduction to Probability Theory

Bayesian Probabilities

- We have viewed probabilities in terms of the frequencies of random, repeatable events.
- We refer to this as the classical or frequentist interpretation of probability
- Now we turn to the more general Bayesian view, in which probabilities provide a quantification of uncertainty

Introduction to Probability Theory

Bayesian Probabilities

- Consider an uncertain event, for example whether the Arctic ice cap will have disappeared by the end of the century
- This is not a event that can be repeated numerous times
- Our assessment of matters such as the speed the polar ice is melting will affect the actions we take
- We would like to quantify the uncertainty and take optimal actions

Introduction to Probability Theory

Bayesian Probabilities

- Consider the polynomial fitting example, we would like to quantify the uncertainty of the appropriate choice for the parameters $\mathbf{w}$
- We capture our assumptions about $\mathbf{w}$, before observing the data, in the form of a prior probability distribution $p(\mathbf{w})$
- The effect of the observed data $\mathcal{D} = \{t_1, ..., t_N\}$ is expressed through the conditional probability $p(\mathcal{D}|\mathbf{w})$
- With Bayes' theorem, we have

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})}$$

which allows us to evaluate the uncertainty in $\mathbf{w}$ after we observed $\mathcal{D}$

Introduction to Probability Theory

Bayesian Probabilities

- The quantity $p(\mathcal{D}|\mathbf{w})$ can be viewed as a function of $\mathbf{w}$, in which case it is called the likelihood function
- It indicates how probable the observed data is for different settings of the parameters $\mathbf{w}$
- The likelihood is not a probability distribution over $\mathbf{w}$, and its integral with respect to $\mathbf{w}$ does not need to be 1
- To ensure the posterior distribution $p(\mathbf{w}|\mathcal{D})$ is a valid probability density and integrates to one, we integrating both sides of likelihood function with respect to $\mathbf{w}$ and have

$$\int p(\mathbf{w}|\mathcal{D})\,\mathrm{d}\mathbf{w} = \int \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})}\,\mathrm{d}\mathbf{w} = 1 \implies p(\mathcal{D}) = \int p(\mathcal{D}|\mathbf{w})p(\mathbf{w})\,\mathrm{d}\mathbf{w}$$

Introduction to Probability Theory

Bayesian Probabilities

- From the Bayesian viewpoint, there is only a single data set $\mathcal{D}$ (the observed one), and the uncertainty in the parameters is expressed through a probability distribution over $\mathbf{w}$
- A widely used estimator is maximum likelihood, in which $\mathbf{w}$ is set to the value that maximizes the likelihood function
- In the machine learning literature, the negative log of the likelihood function is called an error function
- Because the negative logarithm is monotonically decreasing, maximizing the likelihood is equivalent to minimizing the error

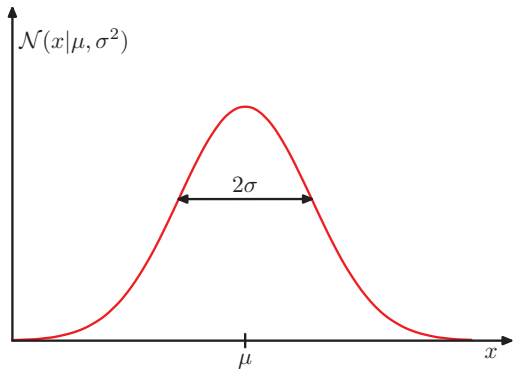
Introduction to Probability Theory

Gaussian distribution

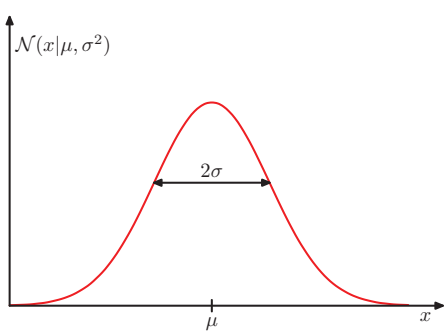
- The normal or Gaussian distribution for a single real-valued variable x, with mean μ and variance σ^2, is given by:

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

where σ is called the standard deviation, and $\beta = 1/\sigma^2$, is called the precision



Introduction to Probability Theory



- From the definition, the Gaussian distribution satisfies

$$\mathcal{N}(x|\mu,\sigma^2) > 0$$

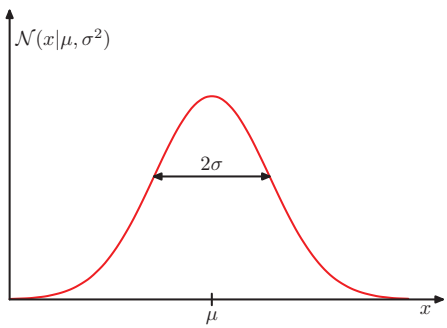
$$\int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2)\,\mathrm{d}x = 1$$

Thus, it is a valid probability density

- We can readily find expectations of functions of x under the Gaussian distribution

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2)x\,\mathrm{d}x = \mu$$

Introduction to Probability Theory



- Similarly, for the second order moment

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} \mathcal{N}(x|\mu,\sigma^2)x^2\,\mathrm{d}x = \mu^2 + \sigma^2$$

- With that, the variance of x is given by

$$\mathrm{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \sigma^2$$

- The maximum of a distribution is known as its mode. For a Gaussian, the mode coincides with the mean

Introduction to Probability Theory

Python Code for Normal Distribution

```python
# Python Program for generating data from a normal distribution
# Load required lib
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# define function to generate data
def generate_normal_data(mean=0, std_dev=1, size=1000):
    data = np.random.normal(mean, std_dev, size)
    return data

# define function for plotting the generated data
def plot_data(data, mean, std_dev, bins=30):
    ...
```

Introduction to Probability Theory

Python Code for Normal Distribution

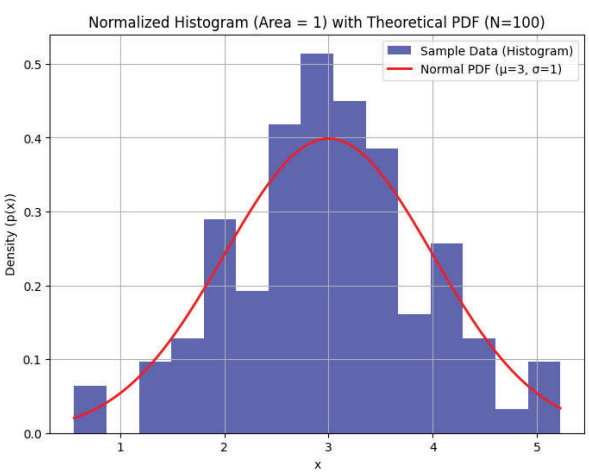
```
# ... Continue from previous page ...

# Example usage
np.random.seed(42)
mean = 10
std_dev = 2
size = 100
data = generate_normal_data(mean, std_dev, size)
plot_data(data, mean, std_dev)
```


Introduction to Probability Theory

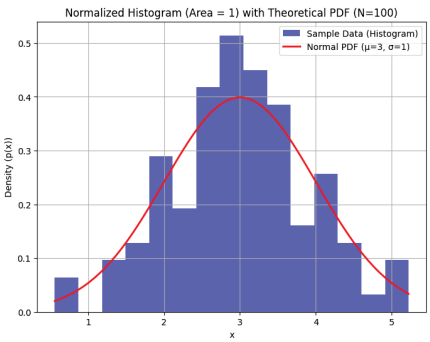
Data from Normal Distribution

- Example: we have a data set of N observations of variable x, $\mathbf{x} = (x_1, \ldots, x_N)^{\mathrm{T}}$




```
[3.03122328 2.89930654 3.02843662 3.45986408 3.65849396 2.93250205
 2.03084223 2.59064499 1.89587608 0.55731949 2.98780229 1.96393714
 3.13997467 3.89368931 3.33675765 3.14151629 2.12900797 2.48342034
 3.09928788 4.34013134 3.19202319 4.15491066 2.63971744 3.26016519
 2.99521194 2.74556237 3.6655147  2.40160154 4.02039936 3.98817904
 4.2231845  2.59222109 2.63083015 4.47874509 1.85938673 3.62363483
 1.69415306 3.19849853 2.94979207 1.99679865 4.01580043 3.0407985
 2.17485868 4.05152556 2.83383093 2.4840089  3.63042288 4.11352111
 0.7936058  2.23031802 4.30006324 3.86321451 1.89463443 2.95380039
 3.66387521 2.66569981 1.9499318  1.3209125  1.72292804 2.46282483
 3.24954285 1.94880807 4.12742982 3.05790988 3.44112268 3.4865838
 5.22417193 3.7448373  3.3772082  3.31292654 4.88021615 3.8128536
 2.33145187 3.33985125 2.51604532 5.09409513 2.60498208 2.98221227
 3.13975644 3.35784106 1.88323448 2.50928847 2.97834113 2.15278597
 3.59569438 2.93995473 2.88955459 3.18968742 1.67156252 2.70844526
 2.6715849  1.18352512 3.10414987 1.27317256 1.76568527 4.9334422
 3.38052029 3.85319669 4.55617993 3.00254927]
```

Introduction to Probability Theory



- Suppose that the observations are drawn independently, we would like to determine Gaussian distribution parameters from the data
- Data points that are drawn independently from the same distribution are said to be independent and identically distributed (i.i.d.)
- The joint probability of two independent events is given by the product of the marginal probabilities for each event separately

Introduction to Probability Theory

- Because our data x is i.i.d., we obtain the probability of the data set,

$$p(\mathbf{x}|\mu, \sigma^2) = \prod_{n=1}^{N} \mathcal{N}(x_n|\mu, \sigma^2)$$

- When viewed as a function of μ and σ^2, this is the likelihood function for the Gaussian distribution
- One common criterion for determining the parameters is to find the parameter values that maximize the likelihood function

Introduction to Probability Theory

- In practice, we often maximize the log of the likelihood function

$$\ln p(\mathbf{x}|\mu, \sigma^2) = \ln \left(\prod_{n=1}^{N} \mathcal{N}(x_n|\mu, \sigma^2) \right)$$

- Recall the Gaussian distribution is given by:

$$\mathcal{N}(x_n|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_n - \mu)^2}{2\sigma^2} \right]$$

- Substituting into the log likelihood yields

$$\ln p(\mathbf{x}|\mu, \sigma^2) = \ln \left\{ \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^N \exp\left[-\frac{1}{2\sigma^2} \sum_{n=1}^{N} (x_n - \mu)^2 \right] \right\}$$

Introduction to Probability Theory

- Simplifying the log likelihood with $\ln(ab) = \ln(a) + \ln(b)$, we obtain

$$\ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^{N} (x_n - \mu)^2 - \frac{N}{2} \ln \sigma^2 - \frac{N}{2} \ln(2\pi)$$

- To maximize it with respect to μ, we use the standard rules of differentiation

$$\frac{\partial}{\partial \mu} \ln p(\mathbf{x}|\mu, \sigma^2) = \frac{1}{\sigma^2} \sum_{n=1}^{N} (x_n - \mu)$$

- Setting this equal to zero yields

$$\sum_{n=1}^{N} x_n = N\mu$$

Introduction to Probability Theory

- From that, we obtain the maximum likelihood solution μ_{ML}

$$\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

- Now, we use the same strategy to determine σ^2
- Taking the derivative of the likelihood with respect to σ^2, we obtain

$$\frac{\partial}{\partial\sigma^2}\ln p(\mathbf{x}|\mu,\sigma^2) = \frac{1}{2(\sigma^2)^2}\sum_{n=1}^{N}(x_n-\mu)^2 - \frac{N}{2}\frac{1}{\sigma^2}$$

Introduction to Probability Theory

- We set the derivative to zero and obtain

$$\frac{N}{2}\frac{1}{\sigma^2} = \frac{1}{2(\sigma^2)^2}\sum_{n=1}^{N}(x_n-\mu)^2$$

- From that, we obtain the maximum likelihood solution for σ^2_{ML}

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}(x_n-\mu_{\mathrm{ML}})^2$$

- Thus, once we found μ_{ML}, we can use that to compute σ^2_{ML}

Introduction to Probability Theory

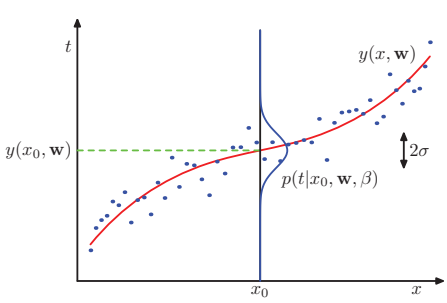
- Now for the polynomial fitting problem, we wish to make predictions for the target variable t given some new value of the variable x
- We can express our uncertainty over the value of the target variable t using a probability distribution
- Given a new value of x, we assume the corresponding value of t has a Gaussian distribution with a mean equal to $y(x, \mathbf{w})$

$$p(t|x, \mathbf{w}, \beta) = \mathcal{N}(t|y(x, \mathbf{w}), \beta^{-1})$$

- Recall the polynomial function $y(x, \mathbf{w})$ is defined as

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \cdots + w_M x^M$$

Introduction to Probability Theory



- Now, we use the training data $\{\mathbf{x}, \mathbf{t}\}$ to determine the values of the unknown parameters $\mathbf{w}$ and β
- If the data are assumed to be drawn independently from the distribution, then

$$p(\mathbf{t}|\mathbf{x}, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|y(x_n, \mathbf{w}), \beta^{-1})$$

- The log of the likelihood function is

$$\ln p(\mathbf{t}|\mathbf{x}, \mathbf{w}, \beta) = -\frac{\beta}{2} \sum_{n=1}^{N} \left(y(x_n, \mathbf{w}) - t_n\right)^2 + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi)$$

Introduction to Probability Theory

- Maximizing the likelihood with respect to $\mathbf{w}$ is the same as minimizing the negative log likelihood

$$\text{minimizing:} \quad \frac{\beta}{2}\sum_{n=1}^{N}\left(y(x_n,\mathbf{w})-t_n\right)^2$$

- Thus, maximizing likelihood is equivalent, so far as determining $\mathbf{w}$ is concerned, to minimizing the sum-of-squares error function
- To find the precision parameter β, we maximize log likelihood with respect to β

$$\frac{1}{\beta_{\mathrm{ML}}}=\frac{1}{N}\sum_{n=1}^{N}\left(y(x_n,\mathbf{w}_{\mathrm{ML}})-t_n\right)^2$$

- We can first determine $\mathbf{w}_{\mathrm{ML}}$ and subsequently use it to find β_{ML}

Introduction to Probability Theory

- Having determined the parameters $\mathbf{w}$ and β, we can now make predictions for new values of x

$$p(t|x,\mathbf{w}_{\mathrm{ML}},\beta_{\mathrm{ML}})=\mathcal{N}\left(t|y(x,\mathbf{w}_{\mathrm{ML}}),\beta_{\mathrm{ML}}^{-1}\right)$$

- Now we introduce a prior distribution over $\mathbf{w}$ and consider a Gaussian distribution of the form

$$p(\mathbf{w}|\alpha)=\mathcal{N}(\mathbf{w}|\mathbf{0},\alpha^{-1}\mathbf{I})=\left(\frac{\alpha}{2\pi}\right)^{(M+1)/2}\exp\left\{-\frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}\right\}$$

  where α is the precision of the distribution, and $M+1$ is the total number of elements in the vector $\mathbf{w}$
- Variables such as α, which control the distribution of model parameters, are called hyperparameters.

Introduction to Probability Theory

- Using Bayes' theorem, the posterior distribution for $\mathbf{w}$ is proportional to the product of the prior distribution and the likelihood function

$$p(\mathbf{w}|\mathbf{x}, \mathbf{t}, \alpha, \beta) \propto p(\mathbf{t}|\mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w}|\alpha)$$

- Taking the negative logarithm, we find the maximum of the posterior is given by the minimum of

$$\frac{\beta}{2}\sum_{n=1}^{N}\left\{y(x_n, \mathbf{w}) - t_n\right\}^2 + \frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}$$

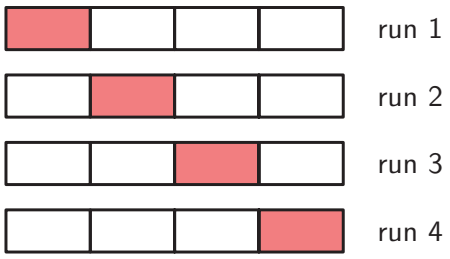
- Thus we see that maximizing the posterior distribution is equivalent to minimizing the regularized sum-of-squares error function encountered earlier ($\lambda = \alpha/\beta$)

Introduction to Probability Theory

Model Selection

- In the polynomial curve fitting example, we saw that there was an optimal order of polynomial.
- The order of the polynomial governs the model complexity
- With regularized least squares, the regularization coefficient λ also controls the effective complexity of the model
- For more complex models, such as neural networks, there will be multiple parameters governing complexity.
- Sometimes, we may wish to consider a range of different types of models in order to find the best one for our application

Introduction to Probability Theory



- In many applications, the data for training and testing will be limited
- In order to build good models, we wish to use as much of the available data as possible for training.
- However, if the validation set is small, it will give a relatively noisy estimate of predictive performance.
- One solution to this dilemma is to use cross-validation

Outline

- Introduction
- An Example of Supervised Learning
- Introduction to Probability Theory
- Introduction to Decision Theory
- Introduction to Information Theory

Introduction to Decision Theory

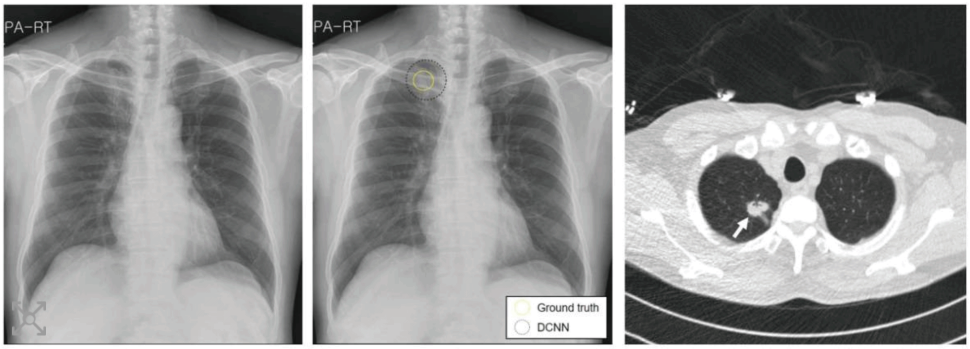
- Decision theory, when combined with probability theory, allows us to make optimal decisions in situations involving uncertainty
- Suppose we have an input vector $\mathbf{x}$ together with a corresponding target vector $\mathbf{t}$, and our goal is to predict $\mathbf{t}$ given a new value for $\mathbf{x}$
- The joint probability distribution $p(\mathbf{x}, \mathbf{t})$ provides a complete summary of the uncertainty associated with these variables
- Determination of $p(\mathbf{x}, \mathbf{t})$ from training data is an example of inference
- Practically, we must often take a specific action based on what value $\mathbf{t}$ is likely to take, and this aspect is the subject of decision theory

Introduction to Decision Theory

- Consider a medical diagnosis problem. We wish to determine whether a patient has cancer or not from X-ray image

Deep learning helps radiologists detect lung cancer on chest X-rays

13 Nov 2019 Tami Freeman



Introduction to Decision Theory

- The input vector $\mathbf{x}$ is a set of pixel intensities in the image, and output variable t represents the presence of cancer $\mathcal{C}_1$, or the absence of cancer $\mathcal{C}_2$
- The general inference problem involves determining the joint distribution $p(\mathbf{x}, \mathcal{C}_k)$
- In the end, we must decide either to give treatment to the patient or not, and we would like this choice to be optimal
- This is the decision step which will tell us how to make optimal decisions given the appropriate probabilities

Introduction to Decision Theory

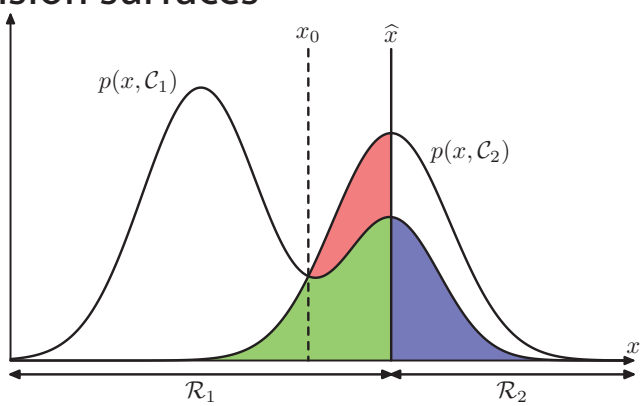
- Consider how we might expect probabilities to play a role in making decisions.
- We are interested in the probabilities of the two classes given the image, which are given by $p(\mathcal{C}_k|\mathbf{x})$.
- Using Bayes' theorem, these probabilities can be expressed as

$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\mathbf{x})}$$

- We can interpret $p(\mathcal{C}_k)$ as the prior probability for the class $\mathcal{C}_k$, and $p(\mathcal{C}_k|\mathbf{x})$ as the corresponding posterior probability.

Introduction to Decision Theory

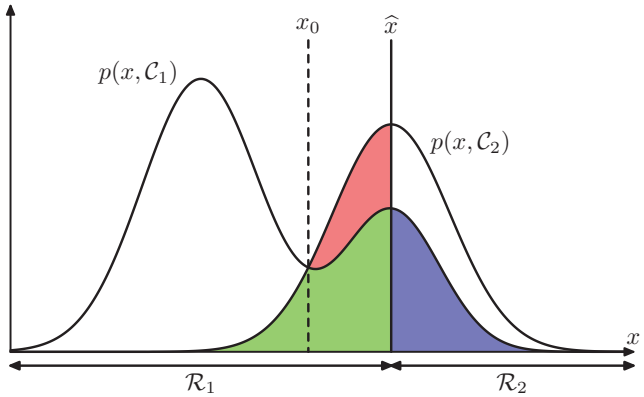
- To reduce misclassifications, we need a rule that assigns each $\mathbf{x}$ to one of the available classes
- This rule will divide the input space into decision regions $\mathcal{R}_k$, one for each class, such that all points in $\mathcal{R}_k$ are assigned to class $\mathcal{C}_k$
- The boundaries between decision regions are called decision boundaries or decision surfaces



Introduction to Decision Theory

- Consider the case of two classes, a mistake occurs when an input vector belonging to class $\mathcal{C}_1$ is assigned to class $\mathcal{C}_2$ or vice versa.
- The probability of this occurring is given by

$$
\begin{aligned}
p(\text{mistake}) &= p(\mathbf{x} \in \mathcal{R}_1, \mathcal{C}_2) + p(\mathbf{x} \in \mathcal{R}_2, \mathcal{C}_1) \\
&= \int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2)\, \mathrm{d}\mathbf{x} + \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1)\, \mathrm{d}\mathbf{x}
\end{aligned}
$$



Introduction to Decision Theory

- In a more general case of K classes, it is slightly easier to maximize the probability of being correct,

$$p(\text{correct}) = \sum_{k=1}^{K} p(\mathbf{x} \in \mathcal{R}_k, \mathcal{C}_k)$$
$$= \int_{\mathcal{R}_k} p(\mathbf{x}, \mathcal{C}_k)\, \mathrm{d}\mathbf{x}$$

- Using the product rule $p(\mathbf{x}, \mathcal{C}_k) = p(\mathcal{C}_k|\mathbf{x})p(\mathbf{x})$, and noting $p(\mathbf{x})$ is common to all terms, we see that $\mathbf{x}$ should be assigned to the class having the largest posterior probability

Introduction to Decision Theory

- If a healthy patient is incorrectly diagnosed as having cancer, the consequences may be some distress and follow-up examinations
- Conversely, if a patient with cancer is diagnosed as healthy, the result may be premature death due to lack of treatment
- Thus, the consequences of these two types of mistake can be dramatically different
- It would be better to make fewer mistakes of the second kind, even if this was at the expense of making more mistakes of the first kind
- We can formalize such issues through a loss function or cost function

Introduction to Decision Theory

- For a new value of $\mathbf{x}$, suppose the true class is $\mathcal{C}_k$ and that we assign $\mathbf{x}$ to class $\mathcal{C}_j$ (where j may or may not be equal to k)
- In so doing, we incur some level of loss, denoted by L_{kj}, which we can be the (k, j) element of a loss matrix
- For instance, in our cancer example, we might have a loss matrix as

$$\begin{array}{cc} & \begin{array}{cc} \text{cancer} & \text{normal} \end{array} \\ \begin{array}{c} \text{cancer} \\ \text{normal} \end{array} & \begin{pmatrix} 0 & 1000 \\ 1 & 0 \end{pmatrix} \end{array}$$

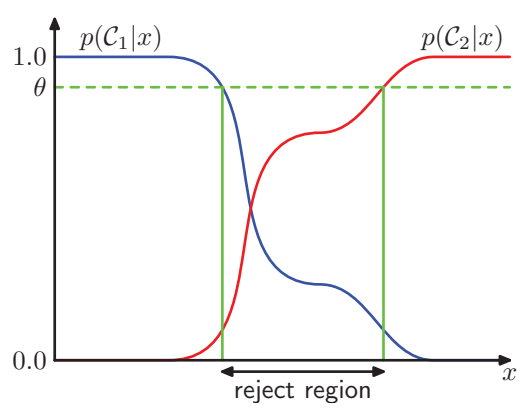
- Our goal is to choose the regions $\mathcal{R}_j$ to minimize the average loss

$$\mathbb{E}[L] = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) \, \mathrm{d}\mathbf{x}$$

Introduction to Decision Theory

Reject Option

- Using the product rule $p(\mathbf{x}, \mathcal{C}_k) = p(\mathcal{C}_k|\mathbf{x})p(\mathbf{x})$ to eliminate the common factor of $p(\mathbf{x})$
- In this case, an automatic system may be suitable for classifying X-ray images for which there is little doubt as to the correct class
- For that, we introduce a threshold θ and rejecting those inputs $\mathbf{x}$ for which the largest of $p(\mathcal{C}_k|\mathbf{x})$ is less than or equal to θ



Introduction to Decision Theory

- Besides classification problems, decision theory can also be applied to regression problems
- Consider the polynomial fitting problem, we would like to estimate the data t with the function $y(x)$
- Suppose we want to choose a specific estimate $y(x)$ for each input x, we incur a loss $L(t, y(x))$. The average loss is

$$\mathbb{E}[L] = \iint L(t, y(x))p(x, t)\,\mathrm{d}x\,\mathrm{d}t$$

Introduction to Decision Theory

- A common choice of loss function in regression problems is the squared loss, so

$$\mathbb{E}[L] = \iint (t - y(x))^2 p(x, t)\,\mathrm{d}x\,\mathrm{d}t$$

- Our goal is to choose $y(x)$ so to minimize $\mathbb{E}[L]$

Outline

- Introduction
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- Introduction to Information Theory

Introduction to Information Theory

- The initial questions treated by information theory lay in the areas of data compression and transmission.
- Consider a discrete random variable x, we ask how much information is received when we observe a specific value for x
- The amount of information can be viewed as the 'degree of surprise' on learning the value of x
- If we are told that a highly improbable event has just occurred, we will have received more information

Introduction to Information Theory

- If we knew that the event was certain to happen we would receive no information
- Our measure of information will depend on the probability $p(x)$
- Find a monotonic function $h(x)$ which represents the information content and depends on the probability $p(x)$
- If we have two events x and y that are unrelated, then the information gain should be the sum of the information gained from each of them,

$$h(x, y) = h(x) + h(y)$$

Introduction to Information Theory

- Two unrelated events will be statistically independent

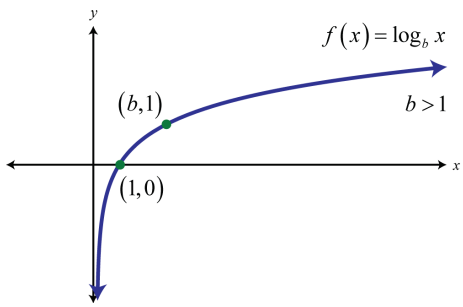
$$p(x, y) = p(x)p(y)$$

- From these two relationships, it is easily shown that $h(x)$ must be a logarithm function of $p(x)$

$$h(x) = -\ln p(x)$$

- The negative sign ensures that information is non-negative.

Introduction to Information Theory



- Low probability events x correspond to high information content.
- The unit of $h(x)$ is bits ('binary digits')
- To transmit the value of x, the average amount of information is defined by the expectation of $h(x)$ with respect to $p(x)$

$$H[x] = -\sum_x p(x) \ln p(x)$$

which is called the entropy of the variable x

Introduction to Information Theory

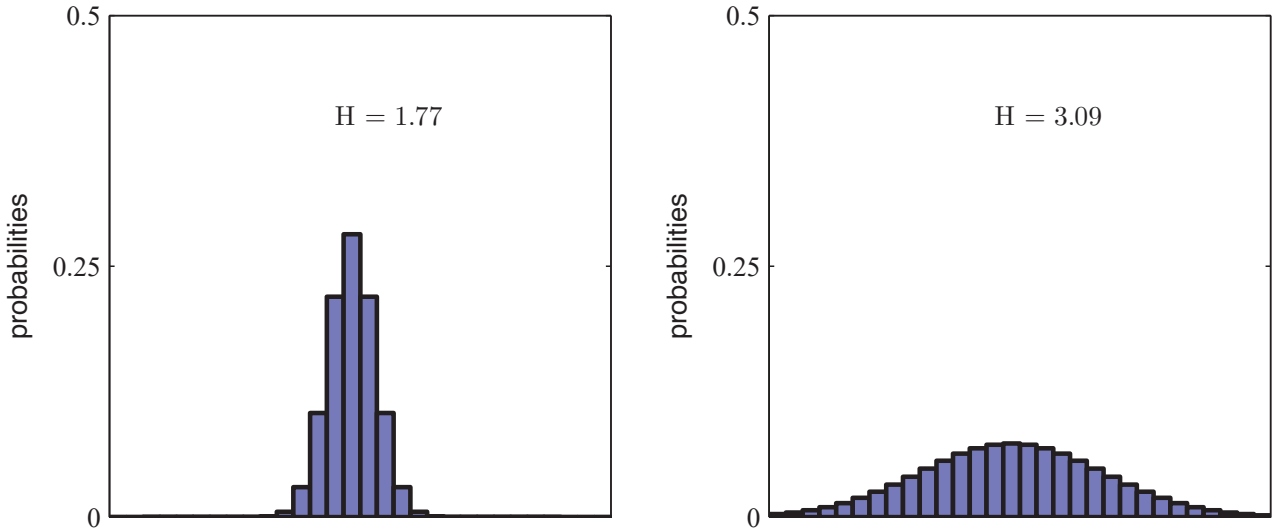
- The entropy of the random variable X is then

$$H[p] = -\sum_i p(x_i) \ln p_i(x_i)$$

- Distributions $p(x_i)$ that are sharply peaked around a few values will have a relatively low entropy,
- Those that are spread more evenly across many values will have higher entropy, why?

Introduction to Information Theory

- Histograms of two probability distributions showing the entropy H is higher when the distribution is broader



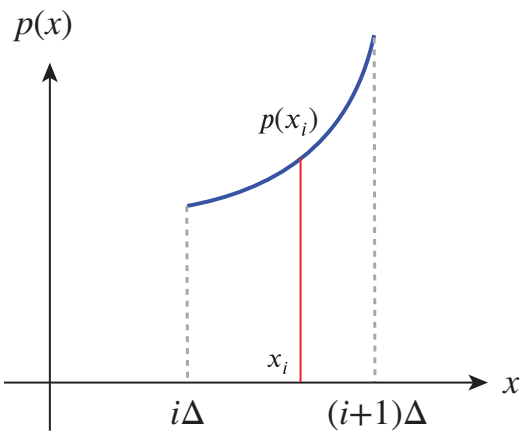
Introduction to Information Theory

- The maximum entropy state can be found by maximizing H using a Lagrange multiplier to enforce the normalization condition

$$\widetilde{H} = -\sum_i p(x_i) \ln p_i(x_i) + \lambda \left(\sum_i p(x_i) - 1 \right)$$

- We found that all of the $p(x_i)$ must be equal in order for $\widetilde{H}$ to be maximum.
- So, $p(x_i) = 1/M$ where M is the total number of x_i
- The corresponding peak value of the entropy is then $H = \ln M$.

Introduction to Information Theory



- Now consider a continuous variable x
- We assume $p(x)$ is also continuous, first divide the domain of x into bins of width Δ
- Then, according to the Mean Value Theorem for Integrals, there is at least one point (x_i) in each bin such that

$$\int_{i\Delta}^{(i+1)\Delta} p(x)\,\mathrm{d}x = p(x_i)\Delta$$

Introduction to Information Theory

- We can now approximate the continuous variable x by x_i whenever x falls in the i-th bin
- The probability of observing the value x_i is then $p(x_i)\Delta$. This gives a discrete distribution for which the entropy becomes

$$H_\Delta = -\sum_i p(x_i)\Delta \ln(p(x_i)\Delta) = -\sum_i p(x_i)\Delta \ln p(x_i) - \ln \Delta$$

- Let $\Delta \to 0$ and omit $\ln \Delta$,

$$\lim_{\Delta\to 0}\left\{-\sum_i p(x_i)\Delta \ln p(x_i)\right\} = -\int p(x)\ln p(x)\,\mathrm{d}x$$

where the quantity on the right-hand side is the differential entropy

Introduction to Information Theory

- In order for the maximum entropy configuration for a continuous variable to be well-defined, we must have

$$\int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1$$
$$\int_{-\infty}^{\infty} xp(x)\,\mathrm{d}x = \mu$$
$$\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\,\mathrm{d}x = \sigma^2$$

- Then, the constrained maximization can be performed using Lagrange multipliers

Introduction to Information Theory

- Next, we maximize the following functional with respect to $p(x)$

$$-\int_{-\infty}^{\infty} p(x)\ln p(x)\,\mathrm{d}x + \lambda_1\left(\int_{-\infty}^{\infty} xp(x)\,\mathrm{d}x - 1\right)$$
$$+\lambda_2\left(\int_{-\infty}^{\infty} xp(x)\,\mathrm{d}x - \mu\right) + \lambda_3\left(\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\,\mathrm{d}x - \sigma^2\right)$$

- Using the calculus of variations, we set the derivative of this functional to zero giving

$$p(x) = \exp\{-1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2\}$$

- Substitute this result into the three constraint equations yields,

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Introduction to Information Theory

- So the distribution that maximizes the differential entropy is the Gaussian
- If we evaluate the differential entropy of the Gaussian, we obtain

$$H[x] = \frac{1}{2}\left\{1 + \ln(2\pi\sigma^2)\right\}$$

- We see again that the entropy increases as the distribution becomes broader